Instructions

- Answer all questions.
- Show all calculations where applicable.
- Draw diagrams whenever required.
- Use proper reasoning instead of writing only definitions.
- Neatly draw Gantt charts for scheduling problems.

1. Explain why an operating system is called both a resource allocator and a control program.
2. Why does every device controller contain a local buffer?
3. Differentiate between Hardware Interrupt vs Software Interrupt.
4. Suppose an operating system had no timer. Discuss at least five problems that would occur.
5. Why are privileged instructions executed only in kernel mode?
6. Differentiate between
    - Process
    - Program
    - Thread
7. Why is PCB considered essential for context switching?
8. Consider the following scenario.

    A parent process creates three child processes.

    - Child 1 completes successfully.
    - Child 2 never terminates.
    - Child 3 finishes but the parent never calls wait().

    Answer:

    a) Which process becomes Zombie?

    b) Which becomes Orphan?

9. What is Cascading Termination?
10. Compare the following IPC methods.
    - Shared Memory
    - Message Passing
11. Differentiate between
    - Direct Communication
    - Indirect Communication
12. Compare
    - User-Level Threads
    - Kernel-Level Threads

13. Differentiate among
    - Many-to-One
    - One-to-One
    - Many-to-Many
    - Two-Level Thread Models

    State which operating systems use each model.

14. FCFS

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Calculate

- Gantt Chart
- Completion Time
- Turnaround Time
- Waiting Time
- Average Waiting Time
- Average Turnaround Time

15. SRTF

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	2
P4	3	1

Calculate

- Gantt Chart
- Completion Time
- Turnaround Time
- Waiting Time

- Average Waiting Time

16. RR

    Time Quantum = 3 ms

Process	Arrival Time	Burst Time
P1	0	10
P2	1	5
P3	2	8
P4	3	6

Calculate all scheduling parameters.

17. Write shell commands for the following questions
    a. Create three directories named Week1, Week2, and Week3 using a single command.
    b. Create two files and merge their contents into a third file named combined.txt.
    c. Display the login name of the current user.
    d. Calculate the value of (25 × 8) + 100 using bc.
    e. Display the line numbers of report.txt, incrementing by 10
    f. Display the contents of student.txt and count the number of lines.
    g. Search for the word "network" ignoring case and display only the matched words.
    h. Display the contents of report.txt, remove duplicate lines, and count the remaining lines.
    i. Extract the e_id column from employee.csv and count the total number of employees.
    j. Write down all the shell commands you have learned up to the midterm.